

Mathematics Stage 4

Factorising Expressions

Book 5

Version: 260930

Feedback:

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Syllabus Content

MA4-ALG-C-01 generalises number properties to operate with algebraic expressions including expansion and factorisation

Factorise algebraic expressions by identifying numerical and algebraic factors

- Identify and list factors of a single term
- Factorise algebraic expressions using knowledge of factors and finding the highest common numerical factor (HCF)
- Factorise algebraic expressions using knowledge of factors by finding a common algebraic factor, including expressions involving more than 2 terms, and verify the result by expansion

Factorising 1

Review

☐ Expand brackets

a $3(2x + 5)$

b $4(x + 3)$

c $5(2y - 1)$

d $2(3a + 4)$

☐ Simplify terms by dividing by a variable

a $\frac{6x}{3}$

b $\frac{15x}{5}$

c $\frac{8y}{2}$

d $\frac{10a}{5}$

e $\frac{18b}{6}$

f $\frac{21c}{3}$

g $\frac{14x}{2}$

h $\frac{12m}{6}$

☐ Simplify terms by dividing by a variable

a $\frac{6x}{x}$

b $\frac{5y}{y}$

c $\frac{x^2}{x}$

d $\frac{10x}{5x}$

e $\frac{18b}{6b}$

f $\frac{21c}{3c}$

g $\frac{14x^2}{2x}$

h $\frac{12m}{6m}$

① Factorising

Factorising is the **inverse** of expanding.

Expanding

 $3(2x + 5) = 6x + 15$

Factorising

To factorise, divide each term by the term outside the brackets.

Put the results inside the brackets.

🔧 Example Factorise, given the factor

Factorise:

$$8y + 12 = 4(\dots\dots\dots)$$

$$= 4\left(\frac{8y}{4} + \frac{12}{4}\right)$$

$$= 4(2y + 3)$$

$$10a - 15 = 5(\dots\dots\dots)$$

$$= 5\left(\frac{10a}{5} - \frac{15}{5}\right)$$

$$= 5(2a - 3)$$

✎ Guided Practice

Factorise

a $6x + 9$

$$= 3(\dots\dots\dots)$$

b $12y + 8$

$$= 4(\dots\dots\dots)$$

c $10a - 5$

$$= 5(\dots\dots\dots)$$

d $14 + 7b$

$$= 7(\dots\dots\dots)$$

e $2xy + 5x$

$$= x(\dots\dots\dots)$$

f $18x^2 - 12x$

$$= 6x(\dots\dots\dots)$$

💡 Key Ideas

1. Factorising is the of expanding.
2. To find what goes inside the brackets, each term on the inside by the term outside the brackets.
3. We can check our answer by our result and confirming we get the original expression.

1 Factorise:

a $4x + 6$
 $= 2(\dots\dots\dots)$

$2(2x + 3)$

b $9y + 12$
 $= 3(\dots\dots\dots)$

$3(3y + 4)$

c $5a + 10$
 $= 5(\dots\dots\dots)$

$5(a + 2)$

d $8 + 12b$
 $= 4(\dots\dots\dots)$

$4(2 + 3b)$

e $7x + 21$
 $= 7(\dots\dots\dots)$

$7(x + 3)$

f $16y + 24$
 $= 8(\dots\dots\dots)$

$8(2y + 3)$

g $6x - 9$
 $= 3(\dots\dots\dots)$

$3(2x - 3)$

h $8y - 12$
 $= 4(\dots\dots\dots)$

$4(2y - 3)$

i $15a - 10$
 $= 5(\dots\dots\dots)$

$5(3a - 2)$

j $20 - 14b$
 $= 2(\dots\dots\dots)$

$2(10 - 7b)$

k $9x - 27$
 $= 9(\dots\dots\dots)$

$9(x - 3)$

l $24y - 16$
 $= 8(\dots\dots\dots)$

$8(3y - 2)$

m $6x + 9y + 12$
 $= 3(\dots\dots\dots)$

$3(2x + 3y + 4)$

n $8a - 12b + 4$
 $= 4(\dots\dots\dots)$

$4(2a - 3b + 1)$

o $10p + 15q - 20$
 $= 5(\dots\dots\dots)$

$5(2p + 3q - 4)$

2 Factorise:

a $6x + xy$
 $= x(\dots\dots\dots)$

$x(6 + y)$

b $y^2 - 12y$
 $= y(\dots\dots\dots)$

$y(y - 12)$

c $6ay + 3a$
 $= 3a(\dots\dots\dots)$

$3a(2y + 1)$

d $6x^2 + 9x$
 $= 3x(\dots\dots\dots)$

$3x(2x + 3)$

e $8y^2 - 12y$
 $= 4y(\dots\dots\dots)$

$4y(2y - 3)$

f $10a^2 + 15a$
 $= 5a(\dots\dots\dots)$

$5a(2a + 3)$

3 In senior years you may need to factor out unusual terms.

a $6x + 7y$
 $= 5(\dots\dots\dots)$

$5\left(\frac{6x}{5} + \frac{7y}{5}\right)$

b $y - 12$
 $= \frac{1}{2}(\dots\dots\dots)$

$\frac{1}{2}(2y - 24)$

c $\pi x + 3$
 $= \pi(\dots\dots\dots)$

$\pi\left(x + \frac{3}{\pi}\right)$

Factorising 2

Review

- ☐ Identify highest common factor

a 6 and 9

b 8 and 10

c 15 and 20

d 12 and 15

e 8, 6, and 10

f 9, 12, and 6

g 3 and 7

h 4 and 8

i $18x$ and 24

j $15y$ and 12

k $16a$ and $12b$

l $5x$ and 15

m 20 and $35a$

n $24y$ and 36

o $14x$ and $28y$

p $9a$ and 15

- ☐ Factorise an expression, given the factor

a $6x + 15 = 3(\dots\dots\dots)$ **b** $4y + 12 = 4(\dots\dots\dots)$ **c** $10a + 15 = 5(\dots\dots\dots)$

Factorising

To factorise:

1. Identify the highest common factor (HCF) of the terms. Place it outside brackets.
2. Divide each term by the HCF. Put the results inside the brackets.

Example Factorise expressions (numerical HCF)

Factorise

$$15a + 10$$

1. $5(\quad)$

2. $5(3a + 2)$

We placed 5 outside the brackets because it is the

..... of $15a$ and 10 .

The inside is $3a + 2$ because we $15a$ and 10 by 5.

$$4a - 12b$$

1. $4(\quad)$

2. $4(a - 3b)$

If we wrote: $4a - 12b = 2(2a - 6b)$, this is not fully factorised because

Guided Practice

Factorise

a $5x + 15$

b $8y + 12$

c $14y - 21$

d $12x + 18y$

e $15 + 9b$

f $6x + 9y + 12$

Key Ideas

1. To factorise an expression:
 - a. Identify the of all terms.
 - b. Write the HCF the brackets.
 - c. each term by the HCF.
 - d. Write the results the brackets.
2. We can check our answer by our result and confirming we get the original expression.

1 Factorise the following.

a $3x + 6$ $= 3(\quad)$ $3(x + 2)$	b $8x + 40$ $= 8(\quad)$ $8(x + 5)$	c $15x - 35$ $5(3x - 7)$	d $5j - 20$ $5(j - 4)$
a $9b - 15$ $3(3b - 5)$	b $12 + 16f$ $4(3 + 4f)$	c $5d - 30$ $5(d - 6)$	d $48 + 12w$ $12(4 + w)$
e $4x + 6$ $2(2x + 3)$	f $9y + 12$ $3(3y + 4)$	g $10a - 15$ $5(2a - 3)$	h $14x - 21$ $7(2x - 3)$
i $18 - 27a$ $9(2 - 3a)$	j $20b + 15$ $5(4b + 3)$	k $14x - 21y + 7$ $7(2x - 3y + 1)$	l $8a - 12b + 4$ $4(2a - 3b + 1)$

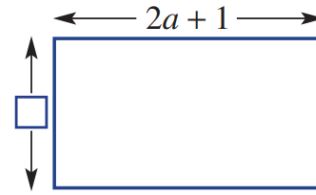
2 Factorise these expressions, if possible.

a $30x + 18$ $6(5x + 3)$	b $30x - 18$ $6(5x - 3)$	c $30x - 19$ can't factorise	d $30x - 20$ $10(3x - 2)$
e $30x + 20$ $10(3x + 2)$	f $35x + 21$ $7(5x + 3)$	g $30x + 18$ $6(5x + 3)$	h $27x + 18$ $9(3x + 2)$
i $18x + 18$ $18(x + 1)$	j $9x + 18$ $9(x + 2)$	k $9x + 19$ can't factorise	l $19x + 19$ $19(x + 1)$

3 Fill in the blanks.

a $3(4x + 1) = \square x + 3$ 12	b $5(7 - 2x) = \square - 10x$ 35
c $6(2 + 5y) = \square + \square y$ 12, 30	d $7(2a - 3b) = \square - \square$ 14a, 21b
e $3(2a + \square) = 6a + 21$ 7	f $4(\square - 2y) = 12 - 8y$ 3
g $7(\square + \square) = 14 + 7q$ 2, q	h $\square(2x + 3y) = 8x + 12y$ 4

- 4 The area of the rectangle is $10a + 5$.
One side's measurement is unknown.
- a What is the value of the unknown measurement?



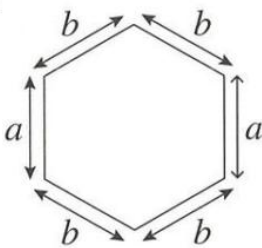
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- b Write an expression for the perimeter of the rectangle.

 $4a + 12$

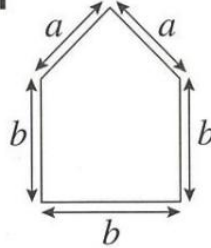
- 5 Match the shapes with the expressions that show their perimeters.

i



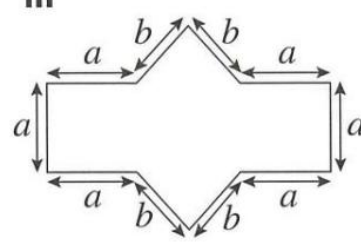
A $2(a + b) + b$

ii



B $2(3a + 2b)$

iii



C $2(a + 2b)$

i-c, ii-A, iii-B

- 6 There are 6 boxes of Smarties and 42 loose Smarties that are not in boxes.
- a Write an expression for the total smarties.

 $6x + 42$

- b Factorise the expression in part a.

 $6(x + 7)$

- c If the smarties are to be shared amongst 6 people, write an expression for the number of Smarties each person receives.

 $x + 7$

- d If the smarties are to be shared amongst 3 people, write an expression for the number of Smarties each person receives.

 $2x + 14$

Factorising 3

Review

- ☐ Identify highest common factor

a $7x$ and $13x$

b $25a$ and $15a$

c $9a$ and $27ab$

d $6xy$ and $12x$

e $2xy$ and $8xy$

f $22xy$ and $44xy$

g $3xr$ and $8r^2$

h $7a$ and $21a^2$

i ab and a^2b

j $3p^2$ and $9pq^2$

k $12qr$ and $8r^2$

l $5x$ and 15

- ☐ Factorise expressions given algebraic factor

a $xy + 5x$
 $= x(\dots\dots\dots)$

b $xy + 6y^2$
 $= y(\dots\dots\dots)$

c $15xy + 10zy$
 $= 5y(\dots\dots\dots)$

- ☐ Factorise expressions by first finding the numerical HCF

a $2x + 6$

b $20 + 5x$

c $12x + 20y^2$

Factorising

To factorise:

1. Identify the highest common factor (HCF) of the terms. Place it outside brackets.
2. Divide each term by the HCF. Put the results inside the brackets.

Example Factorise expressions (algebraic HCF)

Factorise

$$ab - 5b$$

1. $= b(\quad)$

2. $= b(a - 5)$

$$8xy + 12y$$

1. $= 4y(\quad)$

2. $= 4y(2x + 3)$

$$10a^2 - 15a$$

1. $= 5a(\quad)$

2. $= 5a(2a - 3)$

Guided Practice

Factorise

a $xk + xc$

b $x^2 + 3x$

c $x^2 - 3x$

d $6xb + 8b$

e $3ab - 5b$

f $4xy^2 - 6y^2$

Key Ideas

1. To find the highest common factor of two algebraic terms:
 - a. Identify the highest that divides into both terms evenly.
 - b. Identify the variable, and its p..... that is common to both terms.
2. We can check our answer by our result and confirming we get the original expression.

1 Complete the factorisation.

a $2x + 2$ $= 2(\quad)$ $2(x + 1)$	b $5x + 15$ $= 5(\quad)$ $5(x + 3)$	c $x^2 + x$ $= x(\quad)$ $x(x + 1)$
d $3x - 6$ $= 3(\quad)$ $3(x - 2)$	e $x^3 - x^2$ $= x^2(\quad)$ $x^2(x - 1)$	f $8x - 4x^2$ $= 4x(\quad)$ $4x(2 - x)$
g $8y^2 - 12y$ $= 4y(\quad)$ $4y(2y - 3)$	h $12d + 9dz$ $= 3d(\quad)$ $3d(4 + 3z)$	i $15x^3 + 9x^2$ $= 3x^2(\quad)$ $3x^2(5x + 3)$

2 Factorise these expressions.

a $6x + 8$ $2(3x + 4)$	b $6x - 8$ $2(3x - 4)$	c $12x - 16$ $4(3x - 4)$	d $18x - 24$ $6(3x - 4)$
e $9x - 12$ $3(3x - 4)$	f $12 - 9x$ $3(4 - 3x)$	g $12x - 9$ $3(4x - 3)$	h $60x - 45$ $15(4x - 3)$
i $20x - 15$ $5(4x - 3)$	j $20x^2 - 15x$ $5x(4x - 3)$	k $20xy - 15y$ $5y(4x - 3)$	l $20xy - 15x$ $5x(4y - 3)$
m $20 - 15x$ $5(4 - 3x)$	n $21 - 15x$ $3(7 - 5x)$	o $21x - 15xy$ $3x(7 - 5y)$	p $22x - 15xy$ $x(22 - 15y)$

3 Factorise the following.

a $x^2 + 2x$ $x(x + 2)$	b $a^2 - 4a$ $a(a - 4)$	c $y^2 - 7y$ $y(y - 7)$
d $x - x^2$ $x(1 - x)$	e $3p^2 + 3p$ $3p(p + 1)$	f $8x - 8x^2$ $8x(1 - x)$
g $4b^2 + 12b$ $4b(b + 3)$	h $6y - 10y^2$ $2y(3 - 5y)$	i $12a - 15a^2$ $3a(4 - 5a)$
j $9m + 18m^2$ $9m(1 + 2m)$	k $16xy - 48x^2$ $16x(y - 3x)$	l $7xy - 28ab^2$ $7(xy - 4ab^2)$

4 Factorise the following.

a $10cn + 12n$

$2n(5c + 6)$

b $24y + 8ry$

$8y(3 + r)$

c $14jn + 10n$

$2n(7j + 5)$

d $24g + 20gj$

$4g(6 + 5j)$

e $10h + 4z$

$2(5h + 2z)$

f $30u - 20n$

$10(3u - 2n)$

g $40y + 56ay$

$8y(5 + 7a)$

h $12d + 9dz$

$3d(4 + 3z)$

i $21hm - 9mx$

$3m(7h - 3x)$

j $49u - 21bu$

$7u(7 - 3b)$

k $28u - 42bu$

$7u(4 - 6b)$

l $21p - 6c$

$3(7p - 2c)$

5 Fill in the table.

Expanded	Factorised
$3p + 15$	$3(p + 5)$
$3p + 18$	$3(p + 6)$
$6p + 18$	$6(p + 3)$
$6p + 15$	$3(2p + 5)$
$6p + 10$	$2(3p + 5)$
$6p - 10$	$2(3p - 5)$

Expanded	Factorised
$6p^2 - 10p$	$2p(3p - 5)$
$6p^2 - 10$	$2(3p^2 - 5)$
$6 - 10p^2$	$2(3 - 5p^2)$
$-6 + 10p^2$	$-2(3 - 5p^2)$
$-6 + 12p^2$	$6(-1 + 2p^2)$
$-6 + 11p^2$	can't factorise

6 Read the example carefully and answer the questions.

Caroline's work:

$$12x^2 + 8x$$

X

$$12x^2 + 8x$$

$$4(3x^2 + 2x)$$

Why is Caroline's expression not factored completely?

$3x^2 + 2x$ still has a common factor of

Factorise:

$12x^2 - 8x$

$4x(3x - 2)$

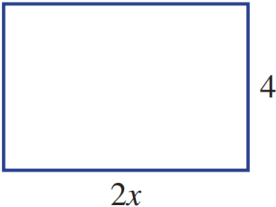
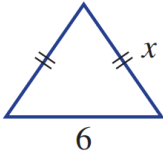
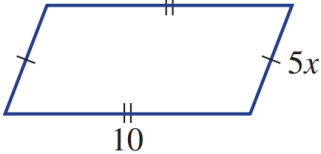
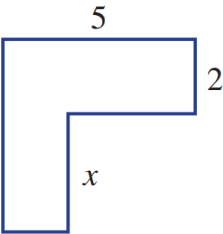
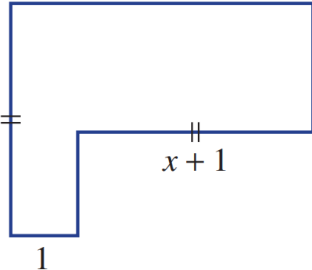
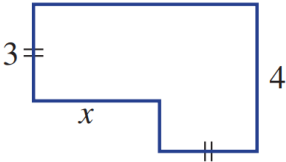
7 Factorise these expressions.

a $6a - 18b + 9c$	b $6a + 18b - 12c$	c $10a + ab$
$3(2a - 6b + 3c)$	$6(a + 3b - 2c)$	$a(10 + b)$
d $10a + ab + a^2$	e $5a^2 + 6ab + 7ac$	f $5pq - 10qr + 15q^2$
$a(10 + b + a)$	$a(5a + 6b + 7c)$	$5q(p - 2r + 3q)$

8 Factorise these expressions.

a $8x - xy$	b $8x - 2xy$	c $8x - 4xy$
$x(8 - y)$	$2x(4 - y)$	$4x(2 - y)$
d $8x - x^2$	e $x^2 - 8x$	f $3x^2 - 12x$
$x(8 - x)$	$x(x - 8)$	$3x(x - 4)$
g $3x^2 - 12x + xy$	h $3x^2 - 12x + 6xy$	i $9x^2 + 3x - 12xy$
$x(3x - 12 + y)$	$3x(x - 4 + 2y)$	$3x(3x + 1 - 4y)$

9 Find the perimeters of these shapes and factorise your answer.

a 	b 	c 
$2(2x + 4)$	$2(x + 3)$	$10(x + 2)$
d 	e 	f 
$2(7 + x)$	$2(2x + 3)$	$2(x + 7)$

10

a	If $x = 5$, work out the value of $2x^2 - 4x$	b	Factorise $2x^2 - 4x$
30		$2x(x - 2)$	
c	Determine the value of x in this equation $2x - 4 = 10$	d	Expand $2x(x + 2)$
$x = 7$		$2x^2 + 4x$	

11 NAPLAN C

A rectangle has an area of $9a^2 - 6a$.
By factorising the expression, find the width of the rectangle if the length is $3a$.

$3a - 2$

12 NAPLAN C

A rectangle has an area of $2pq^2 + 4p^2q$.
Find the length of the rectangle if the width is $2pq$.

$q + 2p$

13 The expression for the area of a rectangle is $(4x^2 + 8x)$.
Find an expression for its width if the length is $(x + 2)$.

$4x$

14 Factorise the following expressions.

a	$3a^2 + 9a + 12$	b	$5z^2 - 10z + zy$
$3(a^2 + 3a + 4)$		$z(5z - 10 + y)$	
c	$x^2 - 2xy + x^2y$	d	$4by - 2b + 6b^2$
$x(x - 2y + xy)$		$2b(2y - 1 + 3b)$	
e	$-12xy - 8yz - 20xyz$	f	$3ab + 4ab^2 + 6a^2b$
$-4y(3x + 2z + 5xz)$		$ab(3 + 4b + 6a)$	

15 These factorisations are incomplete. Complete the factorisation.

a	$2(3x + 9)$	b	$7(4x + 12y)$	c	$x(y + 2xy)$
	$6(x + 3)$		$28(x + 3y)$		$xy(1 + 2x)$
d	$2(6x + 9)$	e	$3x(16zy + 12y)$	f	$12x(x^2 + xy)$
	$6(2x + 3)$		$12xy(4z + 3)$		$12x^2(x + y)$

16 **NAPLAN C**

Fully factorise $24p^2qr^3 + 18p^3q^2 + 6p^2qr^2$

$$6p^2q(4r^3 + 3pq + r^2)$$

17 **NAPLAN C**

Fully factorise $2mn^2 + 6m^2n - 8mn$.

$$2mn(n + 3m - 4)$$

18 Rosie factorised $16xyz + 24xn = 4(4xyz + 6xn)$.

a Explain why she is not fully correct.

The terms in the brackets share common factors.

b Charlotte says it should have been $8(2xyz + 3xn)$. Explain why she is not fully correct.

The terms in the brackets still share a common factor of x .

19 In senior years, it is often necessary to factorise out a fraction. Complete the following:

a	$\frac{x}{2} + \frac{y}{2}$	b	$\frac{3a}{4} - \frac{3b}{4}$	c	$\frac{2p}{3} - \frac{2q}{3}$
	$= \frac{1}{2} (\quad)$		$\frac{3}{4}(a - b)$		$\frac{2}{3}(p - q)$
	$\frac{1}{2}(x + y)$				
d	$\frac{x^2}{3} + \frac{2x}{3}$	e	$\frac{3ab}{5} - \frac{2b}{5}$	f	$\frac{5p^2q}{8} - \frac{3pq}{8}$
	$\frac{x}{3}(x + 2)$		$\frac{b}{5}(3a - 2)$		$\frac{pq}{8}(5p - 3)$

20 Sienna factorised $\frac{5x}{3} + \frac{5y}{6}$ as $\frac{5}{6}(2x + y)$

a By expanding, show that she is correct.

b Hence factorise these expressions; there should be no fractions inside brackets.

i $\frac{x}{2} + \frac{y}{4}$

ii $\frac{3a}{4} - \frac{b}{2}$

iii $\frac{2p}{3} - \frac{q}{6}$

$\frac{1}{4}(2x + y)$

$\frac{1}{4}(3a - 2b)$

$\frac{1}{6}(4p - q)$

iv $\frac{3x}{8} + \frac{y}{4}$

v $\frac{xy^2}{5} + \frac{xy}{10}$

vi $\frac{5c^3}{9} - \frac{cd}{3}$

$\frac{1}{8}(3x + 2y)$

$\frac{xy}{5}(y + 2)$

$\frac{c}{9}(5c^2 - 3d)$

21 Find the value of m if:

$$(x + 3)(m - 2) = -\frac{5x}{2} - \frac{15}{2}$$

$$-\frac{5x}{2} - \frac{15}{2} = -\frac{5}{2}(x + 3)$$

$$\text{So, } m - 2 = -\frac{5}{2}$$

$$x = -\frac{1}{2}$$

22 Find the value of k in each equation.

a $2x(x + 3) + kx(x - 1) = 5x^2 + 4x$

b $ka(a - 2) + 3a(a + 4) = 7a^2 + 6a$

$$\begin{aligned} 2x(x + 3) + kx(x - 1) &= 5x^2 + 4x \\ 2x^2 + 6x + kx^2 - kx &= 5x^2 + 4x \\ (2 + k)x^2 + (6 - k)x &= 5x^2 + 4x \\ \text{Comparing } x^2 \text{ coefficients: } 2 + k &= 5 \rightarrow k = 3 \end{aligned}$$

$$\begin{aligned} ka(a - 2) + 3a(a + 4) &= 7a^2 + 6a \\ ka^2 - 2ka + 3a^2 + 12a &= 7a^2 + 6a \\ (k + 3)a^2 + (12 - 2k)a &= 7a^2 + 6a \\ \text{Comparing } a^2 \text{ coefficients: } k + 3 &= 7 \rightarrow k = 4 \end{aligned}$$